UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

Bachelor's Degree in Aerospace Vehicle Engineering

Euler Project

Mathematics and Computing Engineering

Marina Liao Colom Escolà

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Contents

1 Introduction

Project Euler is a collection of mathematical and computational problems designed to promote problem-solving skills through the application of mathematical reasoning and algorithmic techniques. Many of these problems require combining theoretical results with efficient implementations in order to obtain solutions within reasonable computational limits.

This report presents the resolution of three selected Project Euler problems: Problem 64 (Odd Period Square Roots) [Project Euler, nda], Problem 70 (Totient Permutation) [Project Euler, ndb], and Problem 83 (Path Sum: Four Ways) [Project Euler, ndc]. Although these problems belong to different areas of mathematics and computer science, they share a common requirement: the development of efficient algorithms based on a solid theoretical foundation.

Problem 64 [Project Euler, nda] explores the periodic structure of continued fractions associated with irrational square roots and requires counting the values whose continued fraction expansions have odd period lengths. Problem 70 [Project Euler, ndb] investigates Euler's Totient function and digit permutations, leading to an optimization problem over a large numerical range. Finally, Problem 83 [Project Euler, ndc] addresses a shortest-path problem on a weighted graph derived from a matrix, which is solved using Dijkstra's algorithm [Dijkstra, 1959].

For each problem, the mathematical background is first introduced, followed by the algorithmic approach and its implementation in Python. The obtained results are then discussed and verified against the expected Project Euler solutions.

2 Problem 64

2.1 Description

Project Euler Problem 64 [Project Euler, nda] studies the continued fraction representation of irrational square roots. Every irrational square root can be written as a periodic continued fraction:

$$\sqrt{N} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}}$$

For example,

$$\sqrt{23} = [4; (1, 3, 1, 8)]$$

where the sequence $(1, 3, 1, 8)$ repeats indefinitely.

The objective is to determine how many continued fractions for

$$N \leq 10000$$

have an odd period length.

2.2 Resolution

2.2.1 Mathematical Background

For a non-square integer N, define

$$a_0 = \lfloor \sqrt{N} \rfloor.$$

The continued fraction coefficients are generated using:

$$m_{k+1} = d_k a_k - m_k$$

$$d_{k+1} = \frac{N - m_{k+1}^2}{d_k}$$

$$a_{k+1} = \left\lfloor \frac{a_0 + m_{k+1}}{d_{k+1}} \right\rfloor.$$

The sequence becomes periodic, and for square roots the period ends when

$$a_k = 2a_0.$$

The number of iterations required to reach this condition is the period length.

2.2.2 Algorithmic Approach

The algorithm proceeds as follows:

1. Iterate through all integers from 2 to 10000.
2. Skip perfect squares because their square roots are rational.
3. Compute the continued fraction period using the recurrence relations.
4. Determine whether the period length is odd.
5. Count all values with odd periods.

For each value of N, the period length is relatively small.

The algorithm performs:

$$O(10000 \times P)$$

operations, where P is the average period length.

This is efficient enough to run in less than a second on modern computers.

2.2.3 Python Implementation

```python
import math

def period_length(n):
    a0 = int(math.isqrt(n))

    if a0 * a0 == n:
        return 0

    m = 0
    d = 1
    a = a0
    period = 0

    while True:
        m = d * a - m
        d = (n - m * m) // d
        a = (a0 + m) // d

        period += 1

        if a == 2 * a0:
            break

    return period


def count_odd_periods(limit):
    count = 0

    for n in range(2, limit + 1):
        if period_length(n) % 2 == 1:
            count += 1

    return count


result = count_odd_periods(10000)
print(result)
```

2.3 Result

Running the program produces:

$$1322$$

Therefore, there are

$$\boxed{1322}$$

continued fractions for $N \leq 10000$ whose period length is odd.

3 Problem 70

3.1 Description

The goal of Project Euler Problem 70 [Project Euler, ndb] is to find a number $n < 10^7$ such that:

- $\phi(n)$ is a permutation of n
- The ratio $\frac{n}{\phi(n)}$ is minimized

where $\phi(n)$ is Euler's Totient function, defined as the number of integers less than n that are coprime to n.

3.2 Resolution

3.2.1 Mathematical Background

Euler's Totient function is defined as:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

where the product runs over all distinct prime factors of n.

We search for numbers where n and $\phi(n)$ contain the same digits.

3.2.2 Algorithmic Approach

The algorithm follows these steps:

1. Compute $\phi(n)$ for all $n < 10^7$ using a sieve method.
2. For each n, check if n and $\phi(n)$ are permutations of each other.
3. Compute the ratio $\frac{n}{\phi(n)}$.
4. Track the number that minimizes this ratio.

The sieve computation runs in:

$$O(n \log \log n)$$

The permutation check for each number is:

$$O(d \log d)$$

where d is the number of digits (at most 7 here).

Overall complexity is manageable for $n = 10^7$.

3.2.3 Python Implementation

```python
import math

LIMIT = 10_000_000

def sieve_totients(n):
    phi = list(range(n))

    for i in range(2, n):
        if phi[i] == i:
            for j in range(i, n, i):
                phi[j] -= phi[j] // i
    return phi


```

```python
def is_permutation(a, b):
    return sorted(str(a)) == sorted(str(b))


def solve():
    phi = sieve_totients(LIMIT)

    best_n = 0
    best_ratio = float('inf')

    for n in range(2, LIMIT):
        phi_n = phi[n]

        if is_permutation(n, phi_n):
            ratio = n / phi_n
            if ratio < best_ratio:
                best_ratio = ratio
                best_n = n

    return best_n, best_ratio


print(solve())
```

3.3 Result

The final result is:

$$\boxed{n = 8319823}$$

which minimizes $\frac{n}{\phi(n)}$ among all permutation pairs below 10^7.

4 Problem 83

4.1 Description

Project Euler Problem 83 [Project Euler, ndc] asks us to find the minimum path sum from the top-left to the bottom-right of a square matrix (80x80), where movement is allowed in four directions:

- up
- down
- left
- right

Each cell contains a positive integer cost, and the goal is to minimize the total cost of the path.

The matrix used can be consulted at Appendix A.

4.2 Resolution

4.2.1 Mathematical Background

We model the matrix as a weighted graph:

- Each cell is a node
- Edges connect adjacent cells
- Edge weight = value of destination cell

We must compute the shortest path from $(0,0)$ to $(n-1,n-1)$. Being n the matrix size, in this case $n=80$.

4.2.2 Algorithmic Approach

We use Dijkstra's algorithm [Dijkstra, 1959] because:

- All edge weights are non-negative
- We need a global shortest path, not just local decisions

The steps are:

1. Initialize a distance matrix with infinity
2. Set starting cell distance to its value
3. Use a priority queue to always expand the lowest-cost node
4. Relax neighbors in four directions
5. Stop when reaching the bottom-right cell

Each cell is processed once with a priority queue:

$$O(n^2\log(n^2)) = O(n^2\log n)$$

This is efficient for the given constraints.

4.2.3 Python Implementation

```python
import heapq

def read_matrix(filename):
    with open(filename, "r") as f:
        return [
            list(map(int, line.strip().split(",")))
            for line in f
        ]

def min_path_sum(matrix):
    n = len(matrix)
    directions = [(1,0), (-1,0), (0,1), (0,-1)]

    dist = [[float('inf')] * n for _ in range(n)]
    dist[0][0] = matrix[0][0]

    pq = [(matrix[0][0], 0, 0)]

    while pq:
        cost, x, y = heapq.heappop(pq)

        if (x, y) == (n - 1, n - 1):
            return cost

        if cost > dist[x][y]:
            continue

        for dx, dy in directions:
            nx, ny = x + dx, y + dy

            if 0 <= nx < n and 0 <= ny < n:
                new_cost = cost + matrix[nx][ny]

                if new_cost < dist[nx][ny]:
                    dist[nx][ny] = new_cost
                    heapq.heappush(pq, (new_cost, nx, ny))

    return dist[n - 1][n - 1]

matrix = read_matrix("matrix.txt")

result = min_path_sum(matrix)

print("Minimum path sum:", result)
```

4.3 Result

By entering the file containing the matrix 80x80 from Appendix A into the code, we obtain that the minimum path sum is

$$\boxed{425185}$$

5 Conclusions

The three Project Euler problems studied in this report illustrate the importance of combining mathematical analysis with efficient computational methods. Despite their different domains, each problem requires identifying the underlying mathematical structure and translating it into an algorithm capable of handling the specified constraints.

For Problem 64 [Project Euler, nda], the periodic properties of continued fractions for irrational square roots allowed the determination of all values with odd period lengths up to 10 000. Problem 70 [Project Euler, ndb] demonstrated the practical application of Euler's Totient function and sieve-based techniques to efficiently search a large solution space while minimizing a numerical ratio under permutation constraints. Problem 83 [Project Euler, ndc] showed how a matrix can be modeled as a weighted graph and how Dijkstra's algorithm [Dijkstra, 1959] can be used to compute an optimal path when movement is allowed in four directions.

The results obtained match the known solutions to the corresponding Project Euler problems, confirming the correctness of the mathematical reasoning and implementations. Overall, these exercises highlight how concepts from number theory, graph theory, and algorithm design can be combined to solve challenging computational problems efficiently.

6 Bibliography

References

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[Project Euler, nda] Project Euler (n.d.a). Problem 64: Odd period square roots. Available online: https://projecteuler.net/problem=64. Accessed: 2026-06-01.

[Project Euler, ndb] Project Euler (n.d.b). Problem 70: Totient permutation. Available online: https://projecteuler.net/problem=70. Accessed: 2026-06-01.

[Project Euler, ndc] Project Euler (n.d.c). Problem 83: Path sum - four ways. Available online: https://projecteuler.net/problem=83. Accessed: 2026-06-01.

A Matrix 80x80 for Problem 83

```
4445,2697,5115,718,2209,2212,654,4348,3079,6821,7668,3276,8874,4190,3785,2752,9473,7817,9137,496,7338,3434,7152,4355,4552,7917,7827,2460,2350,691,3514,5880,3145,7633,7199,3783,5066,7487,3285,1084,8985,760,872,8609,8051,1134,9536,5750,9716,9371,7619,5617,275,9721,2997,2698,1887,8825,6372,3014,2113,7122,7050,6775,5948,2758,1219,3539,348,7989,2735,9862,1263,8089,6401,9462,3168,2758,3748,5870
1096,20,1318,7586,5167,2642,1443,5741,7621,7030,5526,4244,2348,4641,9827,2448,6918,5883,3737,300,7116,6531,567,5997,3971,6623,820,6148,3287,1874,7981,8424,7672,7575,6797,6717,1078,5008,4051,8795,5820,346,1851,6463,2117,6058,3407,8211,117,4822,1317,4377,4434,5925,8341,4800,1175,4173,690,8978,7470,1295,3799,8724,3509,9849,618,3320,7068,9633,2384,7175,544,6583,1908,9983,481,4187,9353,9377
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510,3556,5377,1406,5721,4946,2635,7847,4251,8293,8281,6351,4912,287,2870,3380,3948,5322,3840,4738,9563,1906,6298,3234,8959,1562,6297,8835,7861,239,6618,1322,2553,2213,5053,5446,4402,6500,5182,8585,6900,5756,9661,903,5186,7687,5998,7997,8081,8955,4835,6069,2621,1581,732,9564,1082,1853,5442,1342,520,1737,3703,5321,4793,2776,1508,1647,9101,2499,6891,4336,7012,3329,3212,1442,9993,3988,4930,7706
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